

MOVIE RATING ANALYSIS PROJECT REPORT

1. PROJECT TITLE

Movie Rating Analysis using Excel, SQL, Power BI.

2. BUSINESS PROBLEM

This project answer questions such as:

- Which genre generated the highest revenue?
- Which genre generated the lowest revenue?
- Which country recorded the highest total rating?
- What is the highest total rating?
- Which genre generate the highest total ratings?

3. PROJECT OBJECTIVE

- To clean the dataset.
- Analyse movie ratings, genres, revenue and release trends using Excel, SQL, Power BI to generate business insights for a movie streaming platform.
- To write the SQL Queries to answer business questions
- Build an interactive POWER BI Dashboard.
- Create DAX measures.
- Generate business insights, recommendations.

4. DATASET

- Movie_ID
- Title
- Genre
- Year
- Runtime
- Rating
- Votes
- Revenue
- Country

5. DATA INFORMATION

ATTRIBUTE	DESCRIPTION
Dataset Name	Movie Rating Dataset

Tool Used	Excel, My SQL Workbench, Power BI
Skills Demonstrated	SQL, Aggregation, CTEs, Window Functions
Visualization Tool	Power BI

6. DATA CLEANING

- Remove Duplicates
- Remove Blank Rows
- Handling missing values
- Replaced Values

7. EXCEL ANALYSIS

a) Formulas

- Average
- Max
- Count
- Sum
- Sumif
- Averageif
- If
- VLOOKUP
- HLOOKUP
- XLOOKUP
- Countifs
- Index
- Index Match
- Averageifs

b) Conditional Formatting

c) Pivot Table

- Total Revenue by country
- Number of movies by country and Genre
- Average votes by country
- Highest revenue by country

d) Pivot Chart

8. SQL Analysis

SQL Queries

- SELECT
- WHERE
- ORDER BY
- DESC
- LIMIT
- ROUND
- AVERAGE
- GROUP BY
- WINDOW FUNCTION
- CTE
- COUNT
- DISTINCT

Q1. Display all movies

```
SELECT * FROM movie_rating
```

Q2. Show only movie title and genre

```
SELECT Title, Genre  
FROM movie_rating;
```

Q3. Find movies with ratings above 8

```
SELECT Title, Rating  
FROM movie_rating  
WHERE rating > 8;
```

Q4. Find movies released after 2015

```
SELECT Title, Year  
FROM movie_rating  
WHERE Year > 2015;
```

Q5. Find the average movie rating

```
SELECT AVG(Rating) AS Average_Rating  
FROM movie_rating;
```

Q6. Find the highest-rated movie

```
SELECT Title, Rating
FROM movie_rating
ORDER BY Rating Desc
LIMIT 1;
```

Q7. Average rating by genre

```
SELECT Genre,
  ROUND(AVG(Rating),2) AS Avg_Rating
FROM movie_rating
GROUP BY Genre
ORDER BY Avg_Rating DESC;
```

Q8. Total revenue generated by each genre

```
SELECT Genre,
  ROUND(SUM(Revenue),2) AS Total_Revenue
FROM movie_rating
GROUP BY Genre
ORDER BY Total_Revenue DESC;
```

Q9. Top 10 movies by revenue

```
SELECT Title, Revenue
FROM movie_rating
ORDER BY Revenue DESC
LIMIT 10;
```

Q10. Average rating by country

```
SELECT Country,
  ROUND(AVG(Rating),2) AS Avg_Rating
FROM movie_rating
GROUP BY Country
ORDER BY Avg_Rating DESC;
```

11. Movies with more than 50,000 votes

```
SELECT Title, Votes
FROM movie_rating
WHERE Votes > 50000
ORDER BY Votes DESC;
```

12. Count the number of unique countries

```
SELECT COUNT(DISTINCT Country) AS Total_Countries  
FROM movie_rating;
```

13. Rank movies by revenue

```
SELECT Title,Revenue,  
       RANK() OVER(ORDER BY Revenue DESC) AS Revenue_Rank  
FROM movie_rating;
```

14. Top 5 genres by average revenue

```
SELECT Genre,  
       ROUND(AVG(Revenue),2) AS Avg_Revenue  
FROM movie_rating  
GROUP BY Genre  
ORDER BY Avg_Revenue DESC  
LIMIT 5;
```

15. Calculate average rating for each genre

```
SELECT Genre,  
       ROUND(AVG(Rating), 2) AS Average_Rating  
FROM Movie_Rating  
GROUP BY Genre  
ORDER BY Average_Rating DESC;
```

16. Find countries with their maximum movie revenue

```
SELECT Country, MAX(Revenue) AS Max_Revenue  
FROM movie_rating  
GROUP BY Country  
ORDER BY Max_Revenue DESC;
```

17. Find the year with the latest movie release

```
SELECT MAX(Year) AS Latest_Year  
FROM movie_rating;
```

18. Find countries with an average movie rating above 5

```
SELECT Country, AVG(Rating) AS Avg_Rating  
FROM movie_rating  
GROUP BY Country
```

HAVING AVG(Rating) > 5;

19. Count the number of unique genres in the dataset

```
SELECT COUNT(DISTINCT Genre) AS Total_Genres  
FROM movie_rating;
```

20. Find all unique countries that produced movies

```
SELECT DISTINCT Country  
FROM movie_rating  
ORDER BY Country;
```

9. POWER BI

- **DAX Measures**

1. Average movie rating

Average rating = AVERAGE(Movie_Rating_Analysis[Rating])

2. Total revenue

Total Revenue = SUM('Movie_Rating_Analysis'[Revenue])

3. Average Revenue?

Average Revenue = AVERAGE(Movie_Rating_Analysis[Revenue])

4. Highest Rating

Highest Rating = MAX(Movie_Rating_Analysis[Rating])

5. Number of Genres

Genre Count = DISTINCTCOUNT(Movie_Rating_Analysis[Genre])

6. Total Votes

Total Votes = SUM(Movie_Rating_Analysis[Votes])

7. Highest Revenue

Highest Revenue = MAX(Movie_Rating_Analysis[Revenue])

8. Total Rating

Total Rating = SUM('Movie_Rating_Analysis'[Rating])

9. Distinct Countries

Distinct countries = `DISTINCTCOUNT('Movie_Rating_Analysis'[Country])`

10. Action genre revenue

Action Revenue =

`CALCULATE(SUM(Movie_Rating_Analysis[Revenue]),Movie_Rating_Analysis[Genre]="Action")`

10. POWER BI DASHBOARD

KPI CARDS

- Total Revenue
- Average Revenue
- Highest Rating
- Genre Count
- Distinct Countries

Charts

- Stacked Bar Chart – Sum of Revenue by Genre
- Line Chart – Average Rating by Year
- Stacked Column Chart – Sum of Rating by Country
- Pie Chart – Count of Country by Genre

Interactive Features

- Year slicer
- Genre Slicer

11. Business Insights

- Thriller movie generated the highest revenue
- Thriller movies contributed 18k in total revenue, making it the highest performing genre
- Drama generated the lowest revenue among the genre
- Drama movies generated 6k in revenue.
- This suggests that Drama movies contributed less to total revenue compared to other genre.
- Japan recorded the highest total rating
- Japan has the highest movie rating among all the countries.

- This indicated stronger audience ratings.
 - The highest movie rating is 8.90
- The dashboard shows that the highest rated movie received a rating of 8.90.
- It indicates the strong audience rating.
 - Thriller movie dominates the rating distribution
- The pie chart shows the thriller approximately 24% of total ratings.

12. Business Recommendations

- Invest more in high performing genres
- Promote highly rated movies

13. Conclusion

This project demonstrate the data analysis using Excel,SQL,Power BI.Through data cleaning, SQL query,DAX Calculation, dashboard.The analysis identified key trends in rating, revenue.

14. Skills Demonstrated

CATEGORY	SKILLS
EXCEL	• Formulas • Conditional Formatting • Pivot Table • Pivot Chart
SQL	• SELECT • WHERE • ORDER BY • DESC • LIMIT • ROUND • AVERAGE • GROUP BY • WINDOW FUNCTION • CTE • COUNT • DISTINCT
POWER BI	• Data Cleaning • Remove Duplicate values • Handle Missing values • Replaced values

	• Remove blank rows • Dashboard • Slicer
DAX	• AVERAGE • SUM • MAX • DISTINCT • COUNT • CALCULATE